

# What Is Left of a Picture:

## Visual Content Is Written Into the Recurrent Memory of a Hybrid Vision-Language Model, Then Discarded Unread

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### Abstract

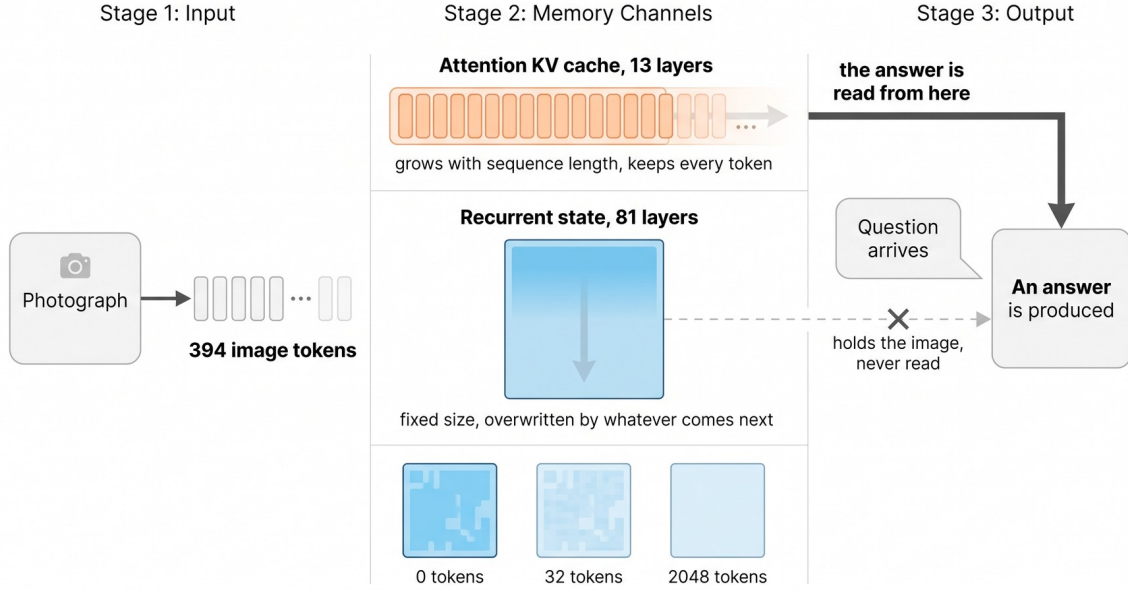
In a transformer, an image persists: it becomes hundreds of key-value entries that any later computation can attend to. In a hybrid linear-attention model it does not. Most layers compress what they read into a fixed-size recurrent state that later tokens overwrite. We ask what remains of an image in that state, and whether the model uses it. Working with Zamba2-VL-7B, whose 81 decoder layers all carry a Mamba2 mixer while only 13 additionally carry attention, we train linear probes on the recurrent state alone and pair them with a causal channel splice. Three findings. First, at the moment of writing the state is highly informative about *which* object is present (83.1% against a 12.5% chance rate,  $p = 6 \times 10^{-92}$ ) and only marginally informative about *how many* (19.4%) or *where* (17.5%): it records gist, not particulars. Second, this content collapses to chance within roughly 32 tokens and never returns, while the model’s own accuracy stays above 95% out to 2048 tokens. Third, the splice explains the discrepancy: exchanging all 81 recurrent states between two runs that saw different images leaves the answer unchanged (17  $\rightarrow$  19 of 30 following the model’s own image), whereas exchanging the 13 attention caches flips it completely (0 of 30 own image, 15 of 30 the other), and exchanging both is indistinguishable from exchanging attention alone. The recurrent memory therefore holds a real, measurable impression of the image that plays no causal role in the answer. Every current mechanistic account of visual hallucination is expressed in terms of attention over image tokens; none of it can describe the six-sevenths of this model where no such attention exists. Total compute: under \$30 of on-demand GPU rental.

## 1 What Was Known Before, and What Is New Here

This section is placed before the results deliberately, so that every claim below can be read against an explicit boundary.

### What was already established

- **Recurrent states in hybrid VLMs accumulate information, and discarded tokens persist in them.** Jiang et al. [1] studied token reduction in hybrid Mamba-Transformer VLMs and introduced a representation-based probing method measuring how much information from discarded tokens is retained, concluding that reduction there behaves “more like compression than dropping.” Their goal is prefill throughput.
- **Vision-language models are sensitive to the order of image and question, and the sensitivity has an attention-based explanation.** Abhinandan et al. [2] identified a “question-first paradox” across five open VLMs, traced it with logit-lens and a causal attention knockout to a question



**Figure 1: The idea.** A photograph becomes 394 image tokens and is written into two memories at once: an attention KV cache that grows with the sequence, present in 13 layers; and a fixed-size recurrent state that is overwritten by whatever follows, present in all 81. The answer is read from the 13. The 81 hold the image and are never read.

“stranded behind hundreds of image tokens,” and resolved it by echoing the question on both sides of the image. Gupta and Gandelsman [3] independently localised the same ordering failure to a narrow mid-network region using activation patching.

- **Visual attributes can be linearly recoverable from hidden states yet unavailable to the model’s own output.** Osaka et al. [4] introduced a Visual Access Sweep over Qwen2.5-VL and InternVL3, defined a Visual Access Boundary, and showed with a probe-versus-decode check that the bottleneck sits at readout rather than at computation.

## What none of that covers, and what we contribute

| Prior work                                                                                                     | Not addressed by it                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Jiang et al. [1]: probes retention of discarded tokens in a hybrid VLM’s recurrent state, for token reduction. | Which <i>visual properties</i> survive; how retention varies with distance from the image; whether the retained content is causally used.                           |
| Abhinandan et al. [2]; Gupta and Gandelsman [3]: image/question ordering and its attention-based mechanism.    | Both study attention-only VLMs (Qwen2.5-VL, InternVL3 and similar). Their mechanism presupposes attention over image tokens and is undefined for a recurrent layer. |
| Osaka et al. [4]: probe-versus-decode gap in transformer VLMs.                                                 | Transformer only. Does not ask whether a non-attentional memory carries content, nor test it causally.                                                              |

Our contributions, none of which appear in the above:

1. The first property-resolved measurement of what an image becomes inside a recurrent state: a gist-versus-particulars gradient at the moment of writing.
2. The first distance-resolved measurement of visual content in such a state, showing collapse within about 32 tokens against flat behavioural accuracy to 2048.
3. The first causal channel splice in a vision-language model that exchanges the recurrent memory and the attention cache independently, establishing that the recurrent contribution is not small but absent from the causal path.
4. A demonstration that “present but causally inert” is a distinct state from the availability/accessibility gap of [4]: there the content is unreadable by the model; here it is unread.

## 2 Significance

**The short version.** A class of models now being deployed for efficiency keeps most of its memory in a form that, for images, holds a real impression that nothing ever reads. Every deployed mitigation for visual hallucination targets the other one-seventh.

**For people building hybrid models.** The efficiency argument for replacing attention layers with recurrent ones assumes the recurrent layers do comparable work. For exact visual recall in this model they do none: the answer survives the total destruction of all 81 recurrent states and does not survive the replacement of 13 attention caches. That is a concrete design signal about where the attention budget matters and where it can be cut, and it is measurable with the instruments released here.

**For people working on hallucination.** Contrastive decoding, attention re-weighting and visual-attention diagnostics are all defined over attention to image tokens. In a model that is six-sevenths recurrent, those tools address a minority of the computation by construction. Our result is not that the recurrent layers cause hallucination; it is that they have been outside the reach of every method, and that they demonstrably contain information which the current toolkit cannot see.

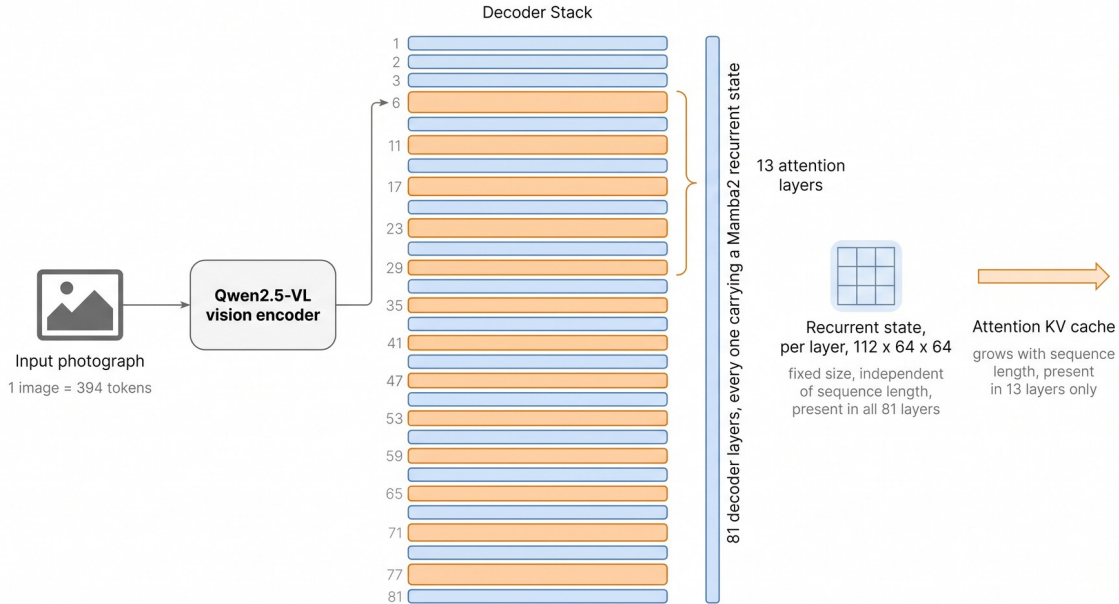
**For interpretability more broadly.** The field’s benchmarks and agenda documents are written for dense transformers. This is a worked example of a finding that only exists in a non-transformer architecture, obtained with instruments that a transformer-only toolkit does not possess, and it argues that architecture coverage is a real gap rather than a stylistic preference.

**What a practitioner should do differently tomorrow.** If you evaluate a hybrid VLM and want to know whether it saw something, do not probe its recurrent state and conclude from a null that the information is absent, nor from a positive that the model is using it. Neither inference is valid in this architecture. Test the channel causally.

## 3 Model and Instruments

**Model.** Zamba2-VL-7B [7] pairs a Zamba2 hybrid language backbone with a Qwen2.5-VL vision encoder. We verified its structure empirically rather than from documentation: 81 decoder layers, *all* of which carry a Mamba2 [5] mixer, of which 13 (indices 6, 11, 17, ..., 77) additionally carry a shared attention block. The recurrent state per layer is [1, 112, 64, 64]; the attention cache exists at those 13

layers only. One image costs 394 tokens. The ratio of recurrent to attention layers is therefore 81:13, more lopsided than the block-type list in the configuration suggests.



**Figure 2: Zamba2-VL-7B, verified by running it.** All 81 decoder layers carry a Mamba2 recurrent state of  $112 \times 64 \times 64$ , fixed in size regardless of sequence length. Thirteen, at the indices marked, additionally carry a shared attention block whose cache grows with the sequence.

**Reading the state.** The Mamba2 recurrence is  $S_t = S_{t-1} \text{diag}(e^{\Delta_t A}) + \Delta_t x_t B_t^\top$  with readout  $y_t = S_t C_t$ , so writing uses  $(x, B)$  and reading uses  $C$ . We hook each mixer’s `in_proj`, whose width 14704 decomposes as  $z(7168) \mid x(7168) \mid B(128) \mid C(128) \mid \Delta(112)$  with two  $B/C$  groups shared across heads. Because a layer’s state has 458,752 entries, we probe a fixed Gaussian random projection to 512 dimensions per layer, which preserves linear separability up to a controlled distortion.

**Probe and controls.** Ridge regression onto one-hot targets, solved in dual form, with cross-validation grouped by block. Every reported number is accompanied by two controls that must land at chance: a **pre-image** control (identical prompt, image never shown) and a **label-permutation** control.

**Splice.** Two runs are prefilled with image and filler only. One channel of the first is then replaced with the second’s, and only then are the question tokens fed through, so the intervention precedes the read. Conditions: no swap, recurrent swapped (all 81), attention swapped (13), and both.

## 4 Materials

Trials are built from COCO val2017 [6] as eight-way forced choices, so chance is exactly 12.5%, in three families spanning coarse to fine: **identity** (which object), **count** (how many), **position** (where, on a  $3 \times 3$  grid with the centre excluded). Trials are constructed in **blocks of eight sharing one identical candidate list**, each candidate correct exactly once, and folds are grouped by block. A previous project

of ours reported a spurious 31.2% against a 12.5% chance rate because trial identity leaked the label; this design forecloses that failure, and the pre-image control is what demonstrates it.

## 5 Results

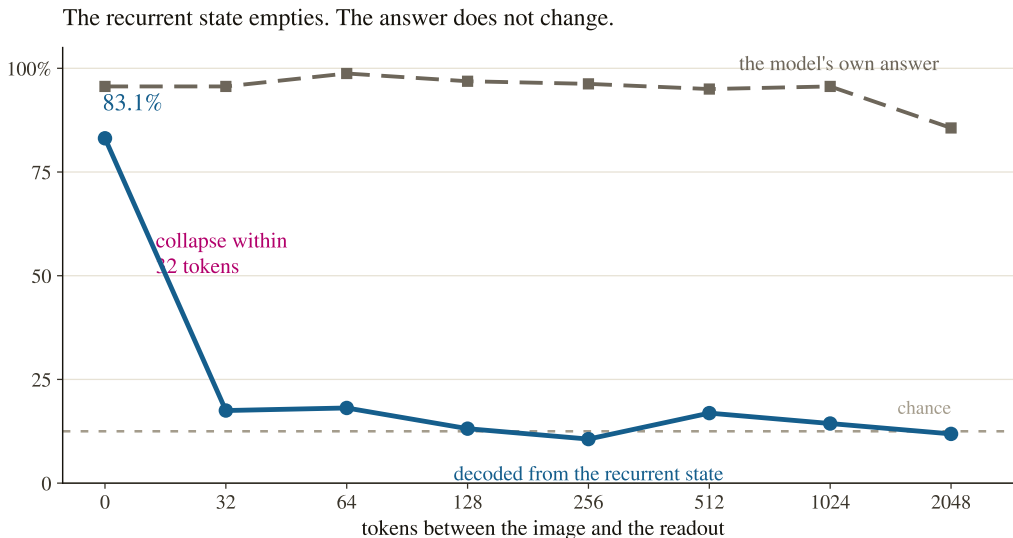
### 5.1 The state records gist, not particulars

|                   | at the write                      | 32 tokens later | model’s accuracy |
|-------------------|-----------------------------------|-----------------|------------------|
| which object      | 83.1% ( $p = 6 \times 10^{-92}$ ) | 17.5%           | 95.6%            |
| how many          | 19.4% ( $p = 0.009$ )             | 11.9%           | 24.4%            |
| where             | 17.5% ( $p = 0.041$ )             | 8.7%            | 47.5%            |
| pre-image control | 12.5%                             | ,               | ,                |

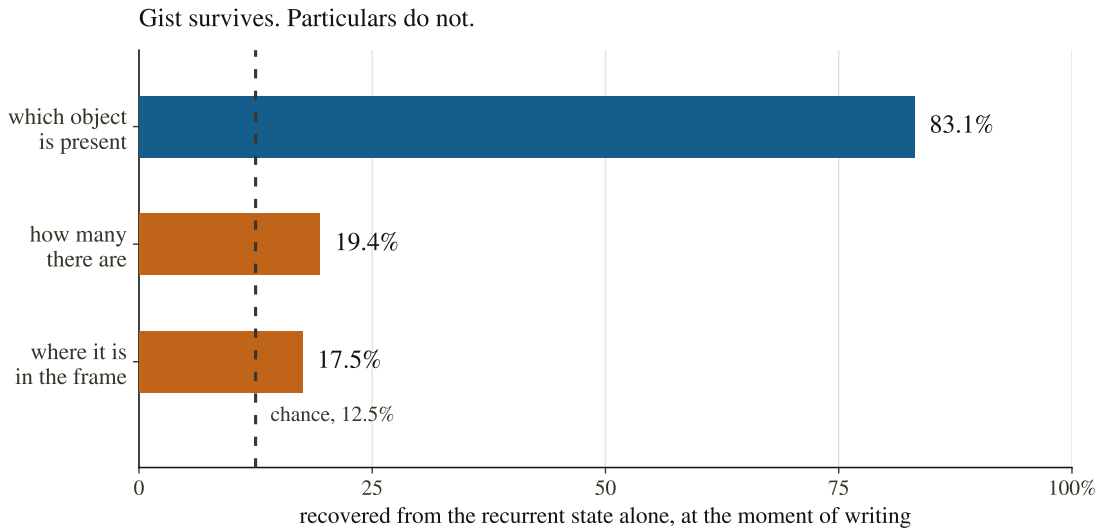
**Table 1:** Chance is 12.5%. The state is highly informative about object identity and barely informative about count or location.

A capacity control rules out the obvious objection to the weak rows. On the identical feature vectors, a coarse scene property (median-split scene complexity) decodes at 71.0% and 74.5% against a 50% chance rate for the count and position families respectively. The features carry visual content in quantity; they do not carry the count.

### 5.2 Collapse within about 32 tokens



**Figure 3:** Probe accuracy against distance from the image. Object identity falls from 83.1% to 17.5% between 0 and 32 tokens and stays near chance out to 2048. The three pre-image controls coincide at 12.5%. This is a step, not an exponential: we deliberately do not quote a half-life, because fitting one to a cliff produces a number (469 tokens) that describes nothing.



**Figure 4:** At the moment of writing, the state is highly informative about which object is present and barely informative about how many or where, from the identical feature vectors.

### 5.3 Gist survives, particulars do not

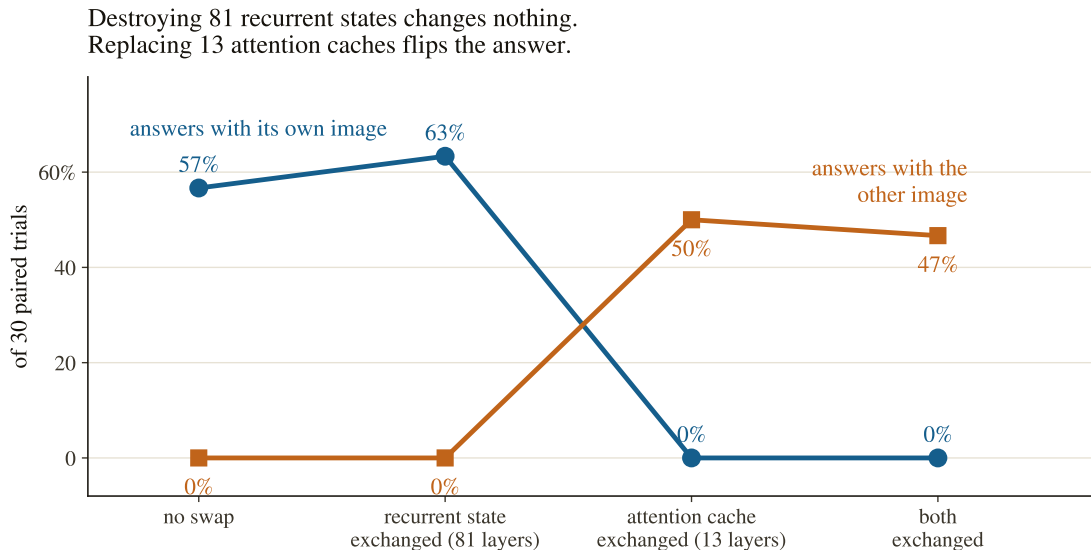
### 5.4 The answer comes from the 13 attention layers

| identity, $n = 30$               | follows own image | follows other image |
|----------------------------------|-------------------|---------------------|
| no swap                          | 17                | 0                   |
| all 81 recurrent states replaced | 19                | 0                   |
| 13 attention caches replaced     | 0                 | 15                  |
| both replaced                    | 0                 | 14                  |

**Table 2:** The two lines cross. Every image is letterboxed to a common canvas so both runs emit identical vision-token counts; zero pairs were dropped for length mismatch, and the recurrent swap was verified to have taken effect in 90/90 checks. That “both” matches “attention” establishes that attention alone determines the answer.

## 6 Limitations

**One model.** Every number here is Zamba2-VL-7B. The architecture class is what makes the question meaningful, but a second hybrid VLM is needed before any claim generalises to the class. **Linear probes.** Absence of linear decodability is not absence of information. The capacity control bounds this concern but does not eliminate non-linear encodings. **Splice baselines.** The model’s own baseline on identity is 17/30 under the letterboxed stimulus, so the flip is measured against a ceiling well below one; count is uninformative for the splice because its baseline is at noise. **A methodological note recorded honestly.** Our first splice forced every image to a square, which equalised token counts but distorted the images and depressed identity accuracy from 17/30 to 8/30. The letterboxed design fixes this. Both runs are in the repository.



**Figure 5:** The causal dissociation as a slope. Exchanging 81 recurrent states leaves the answer with its own image; exchanging 13 attention caches moves it entirely to the other image.

## 7 Conclusion

An image enters this model’s recurrent memory as a vivid impression of what kind of thing it is, loses that impression within about thirty tokens, and none of it matters: the answer was always being read from thirteen attention layers out of eighty-one. The interesting object is not the forgetting. It is that a memory can be measurably full and causally empty at the same time, and that in the architectures now being deployed for efficiency, most of the memory is of that kind.

## References

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- [7] Zephyra. Zamba2-VL-7B model card. <https://huggingface.co/Zephyra/Zamba2-VL-7B>, 2026. Apache-2.0.